

Q1. Unequivocal proof that DNA is the genetic material was first proposed by: {NEET - 2023}

1. Wilkins and Franklin
2. Frederick Griffith
3. Alfred Hershey and Martha Chase
4. Avery, Macleoid and McCarthy

Q2.Expressed Sequence Tags (ESTs) refers to: {NEET - 2023}

1. **Certain important expressed genes.**
2. **All genes that are expressed as RNA.**
3. **All genes that are expressed as proteins.**
4. **All genes whether expressed or unexpressed**

Q3. What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

{NEET - 2023}

1. Transcription of only snRNAs
2. Transcription of rRNAs (28S, 18S and 5.8S)
3. Transcription of tRNA, 5 srRNA and snRNA
4. Transcription of precursor of mRNA

Q4. How many different proteins does the ribosome consist of? {NEET - 2023}

1.	20	2.	80
3.	60	4.	40

Q5. Given below are two statements: {NEET - 2023}

Statement I:	In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.
Statement II:	In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

1. Statement I is incorrect but Statement II is correct.
2. Both Statement I and Statement II are correct.
3. Both Statement I and Statement II are incorrect.
4. Statement I is correct but Statement II is incorrect.



{NEET - 2023}

Choose the correct answer from the options given below:

Q7. Which of the following is not a cloning vector?

1.	Probe	2.	BAC
3.	YAC	4.	pBR322

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In the light of the above statements, choose the correct answer from the options given below:

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A:	Viruses grown in the presence of radioactive phosphorus contained radioactive DNA.
B:	Viruses grown on radioactive sulphur contained radioactive proteins.
C:	Viruses grown on radioactive phosphorus contained radioactive protein
D:	Viruses grown on radioactive sulphur contained radioactive DNA
E:	Viruses grown on radioactive protein contained radioactive DNA

Choose the most appropriate answer from the options given below:

1.	(D) and (E) only	2.	(A) and (B) only
3.	(A) and (C) only	4.	(B) and (D) only

Q17. The salient features of genetic code are:

{NEET - 2023}

- (A) The code is palindromic
(B) UGA act as initiator codon
(C) The code is unambiguous and specific
(D) The code is nearly universal

Choose the most appropriate answer from the options given below:

1.	(A) and (D) only	2.	(B) and (C) only
3.	(A) and (B) only	4.	(C) and (D) only

Q18. Read the following statements and choose the set of correct statements:

{NEET - 2022}

a.	Euchromatin is loosely packed chromatin.
b.	Heterochromatin is transcriptionally active.
c.	Histone octamer is wrapped by negatively charged DNA in nucleosome.



3. Gene mapping

4. Expressed sequence tags

Q22. In an E.coli strain i gene gets mutated and its product cannot bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome ? {NEET - 2022}

1. RNA polymerase will bind the promoter region

2. Only z gene will get transcribed

3. z, y, a genes will be transcribed

4. z, y, a genes will not be translated

Q23. If the length of a DNA molecules is 1.1 metres, what will be the approximate number of base pairs? {NEET - 2022}

1. 6.6×10^6 bp	2. 3.3×10^9 bp
3. 6.6×10^9 bp	4. 3.3×10^6 bp

Q24. Ten E.coli cells with ^{15}N - ds DNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many E.coli cells will have DNA totally free from ^{15}N ? {NEET - 2022}

1. 80 cells

2. 20 cells

3. 40 cells

4. 60 cells

Q25. Which one of the following is not a criterion of genetic material? {NEET - 2022}

1.	Should not provide the scope for changes for evolution
2.	Should be able to express itself in the form of Mendelian character
3.	Should be able to generate its replica
4.	Should be stable chemically and structurally



Q26. DNA replication is semi-conservative in nature was experimentally proved in eukaryotes by:

{NEET - 2022}

1. Hershey and Chase
2. Macleod and McCarty
3. Meselson and Stahl
4. Talyor and his colleagues

Q27. Which one of the following experiments of Frederick Griffith resulted in the discovery of bacterial transformation?

{NEET - 2022}

1.	S-stain(heat-killed) → injected in to Mice → Mice lived
2.	S-strain (heat killed) + R-strain(live) → injected in to Mice → Mice died
3.	S-stain → injected in to Mice → Mice died
4.	R-strain → injected in to Mice → Mice lived

Q28. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

{NEET - 2022}

Assertion (A):	Semiconservative replication was experimentally proved by Mathew Meselson and Franklin Stahl (1958)
Reason (R):	Meselson and Stahl used radioactive isotope ^{15}N and equilibrium density gradient centrifugation technique.

In the light of the above statements, choose the correct answer from the options given below:

1.	(A) is correct but (R) is not correct.
2.	(A) is not correct but (R) is correct.
3.	Both (A) and (R) are correct and (R) is the correct explanation of (A).



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|----|--|
| 4. | Both (A) and (R) are correct but (R) is not the correct explanation of (A) |
|----|--|

Q29. With respect to nucleosome, Which of the following statements is incorrect ?

{NEET - 2022}

1.	Nucleosome contains 120 bp of DNA helix.
2.	Nucleosomes are seen as 'beads' on string' under Electron Microscope.
3.	DNA is wrapped around positively charged histone octamer to form nucleosome.
4.	Nucleosome is the repeating unit of chromatin.

Q30. One of the strands of double stranded DNA has base composition as follows: 15% A, 15% T, 40%G and 30%C. What will be the percentage of these bases in the complementary strand ?

{NEET - 2022}

1. 15% A, 15% T, 30% G , 40% C
2. 15% A, 30% T, 40% G , 15% C
3. 15% A, 15% T, 40% G , 30% C
4. 15% A, 40% T, 15% G , 30% C

Q31. What would be the proportions of light and hybrid density DNA molecule, respectively if Meselson and Stahl's experiment was continued for 60 minutes?

{NEET - 2022}

1. 50%, 50%
2. 25%, 75%
3. 75%, 25%
4. 100%, 0%

Q32. If a DNA molecule is shortened by 25 base pairs, how many helical turns will be reduced from its structure?

{NEET - 2022}

1. 1
2. 3
3. 2.5
4. 2



{NEET - 2022}

- Q34.Match List-I with List- II**

{NEET - 2022}

Choose the correct answer from the options given below :

Q35. Given below are two statements:

{NEET - 2022}



{NEET - 2022}

ence of anticodon on tRNA ?
{NEET - 2022}

{NEET - 2022}

- hRNA $\xrightarrow{(c)}$ (d)
- | |
|----------------------|
| on; (b) -
; |
| on; (d) - Protein |
| ction; (b) - |
| on; (d) - Protein |
| on; (b) -
; |
| ction; (d) - Protein |
| otion; (b) - |
| otion; (d) - |

{NEET - 2021}

	(c) - Trans
4.	(a) - Trans Replication (c) - Trans

{NEET - 2021}

1.	Single nucleotides	2.	Polymorphic DNA
3.	Satellite DNA	4.	Repetitive DNA

{NEET - 2021}

1.	The coding strand in a transcription unit is copied to an mRNA.
2.	Split gene arrangement is characteristic of prokaryotes.
3.	In capping, methylguanosine triphosphate is added to the 3' end of hnRNA.
4.	RNA polymerase binds with the Rho factor to terminate the process of transcription in bacteria.

{NEET - 2021}

1.	T:30 ; G:20 ; C:20	2.	T:20 ; G:25 ; C:25
3.	T:20 ; G:30 ; C:20	4.	T:20 ; G:20 ; C:30

esis of protein? {NEET - 2021}

1. rRNA
2. siRNA
3. mRNA
4. tRNA

{NEET - 2021}

1. DNA Ligase
2. DNase
3. DNA-dependent DNA polymerase
4. DNA-dependent RNA polymerase

{NEET - 2021}

{NEET - 2020}

1.	2.5 meters	2.	2.2 meters
3.	2.7 meters	4.	2.0 meters

{NEET - 2020}

1. Adenine pairs with thymine through one H-bond.
2. Adenine pairs with thymine through three H-bonds.
3. Adenine does not pair with thymine.
4. Adenine pairs with thymine through two H-bonds.

{NEET - 2020}

1.	Polymerases	-	Break the DNA into fragments
2.	Nucleases	-	Separate the two strands of DNA
3.	Exonucleases	-	Make cuts at specific positions within DNA
4.	Ligases	-	Join the two DNA molecules

{NEET - 2020}

1. Recognition of DNA molecule
2. Aminoacylation of tRNA
3. Recognition of an anti-codon
4. Binding of mRNA to ribosome

{NEET - 2020}

- ### 1. Franklin

2. Meischer

3. Chargaff

4. Mendel

Q53. In the polynucleotide chain of DNA, a nitrogenous base is linked to the –OH of:

{NEET - 2020}

1.	2'C pentose sugar	2.	3'C pentose sugar
3.	5'C pentose sugar	4.	1'C pentose sugar

Q54. *E.coli* has only 4.6×10^6 base pairs and completes the process of replication within 18 minutes, then the average rate of polymerization is approximately: {NEET - 2020}

1.	2000 base pairs/second	2.	3000 base pairs/second
3.	4000 base pairs/second	4.	1000 base pairs/second

Q55. Which is the basis of genetic mapping of the human genome as well as DNA fingerprinting?

{NEET - 2020}

1. Polymorphism in the DNA sequence
2. Single nucleotide polymorphism
3. Polymorphism in the hnRNA sequence
4. Polymorphism in the RNA sequence

Q56. Match the following genes of the Lac operon with their respective products:

{NEET - 2019}

(a)	i gene	(i)	β β -galactosidase
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3. Hershey and Chase

4. Jacob and Monod

Q65. In the process of transcription in Eukaryotes, the RNA polymerase I transcribe -
{NEET - 2019}

1. mRNA with additional processing, capping and tailing
2. tRNA, 5 S rRNA and snRNAs
3. rRNAs - 28 S, 18 S and 5.8 S
4. Precursor of mRNA, hnRNA

Q66. What initiation and termination factors are involved in transcription in prokaryotes?
{NEET - 2019}

1. σ and ρ respectively
2. α and β respectively
3. β and γ respectively
4. α and σ respectively



ANSWER KEY

1. (3)	11. (3)	21. (2)	31. (3)	41. (4)	51. (2)	61. (2)
2. (2)	12. (3)	22. (4)	32. (3)	42. (1)	52. (2)	62. (3)
3. (3)	13. (2)	23. (2)	33. (1)	43. (2)	53. (1)	63. (4)
4. (2)	14. (3)	24. (4)	34. (2)	44. (4)	54. (1)	64. (3)
5. (1)	15. (2)	25. (1)	35. (1)	45. (4)	55. (1)	65. (3)
6. (3)	16. (2)	26. (4)	36. (3)	46. (2)	56. (4)	66. (1)
7. (1)	17. (4)	27. (2)	37. (3)	47. (3)	57. (3)	
8. (2)	18. (3)	28. (1)	38. (4)	48. (2)	58. (1)	
9. (4)	19. (2)	29. (1)	39. (1)	49. (4)	59. (2)	
10. (2)	20. (4)	30. (1)	40. (4)	50. (4)	60. (4)	

